

Problem 5

Determine the ratio of:

- (a) The strengths of the gravitational fields,
 - (b) The gravitational potentials,
- at points located at a height $h = 37$ km above the Earth's surface, compared to points on the Earth's surface. The Earth's radius is $R = 6374$ km.

Solution

- (a) The strength of the gravitational field at a distance r from the center of the Earth is given by:

$$G = \frac{GM}{r^2}$$

The ratio of field strengths at height h and on the surface is:

$$\frac{G_h}{G_0} = \frac{R^2}{(R+h)^2}$$

Substitute the values:

$$\frac{G_h}{G_0} = \frac{(6374)^2}{(6374+37)^2} = \frac{40632976}{41186841} \approx 0.987$$

- (b) The gravitational potential at a distance r is given by:

$$\varphi = -\frac{GM}{r}$$

The ratio of potentials is:

$$\frac{\varphi_h}{\varphi_0} = \frac{R}{R+h}$$

Substitute the values:

$$\frac{\varphi_h}{\varphi_0} = \frac{6374}{6374+37} = \frac{6374}{6411} \approx 0.994$$

Conclusion: - The strength of the gravitational field at a height of 37 km is approximately 0.987 of the strength on the surface. - The gravitational potential at that height is approximately 0.994 of the potential on the surface.